Derivations of the Symmetrized Product in a Braided Monoidal Category

Abderrahim Adrabi[a,b]

[a]Faculty of Sciences, Mohammed V University in Rabat, Morocco;
[b]Faculty of Educational Sciences, Mohammed V University in Rabat, Morocco

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ABSTRACT
Let A be an object of a braided monoidal category enriched over abelian groups, and let $\mu\colon A \otimes A \to A$ be a product on A. Let c be the braiding on $A \otimes A$. The symmetrized product of A is $\mu(1+c)$, and the antisymmetrized product is $\mu(1-c)$. A derivation of a product is a morphism $D\colon A \to A$ that satisfies the Leibniz rule for that product. The Leibniz defect of D decides every case treated here. It shows that D is a derivation of $\mu(1 \pm c)$ exactly when the defect composed with $1 \pm c$ vanishes. If no nonzero morphism is killed by 2, a derivation of both products is a derivation of μ, and for a commutative product the derivations of $\mu(1+c)$ and of μ coincide. In a symmetric closed category with kernels the derivations form an object that contains the derivation object of Pareigis. Let α be an endomorphism of A with $\alpha^2 = \mathrm{id}_A$ and let $\sigma = (\alpha \otimes \alpha)c$ be the twist. If μ is commutative and D commutes with α, then a derivation of μ_σ has a defect that is killed by $1 + (\alpha \otimes \alpha)$; if moreover no nonzero morphism is killed by 2, it vanishes on pairs of eigenvectors of α with the same eigenvalue.

KEYWORDS
braided monoidal category; derivation; symmetrized product; Jordan derivation; Leibniz defect

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1. Introduction

A derivation of an algebra A is a linear map D with $D(xy) = D(x)y + xD(y)$. Jacobson [11, Part I] studied the derivations of an arbitrary nonassociative algebra. Every associative algebra carries a second product, the sum $x \circ y = xy + yx$. Jordan, von Neumann and Wigner form a product ab from the two products $a \times b$ and $b \times a$ of an associative system, and call this quasi-multiplication [12, p. 64]. Herstein [9, p. 1104] calls a derivation of this product a Jordan derivation. The sum is twice the Jordan product $\frac{1}{2}(xy + yx)$ of an associative algebra [5, Example 2.1]. Herstein [9, Theorem 3.1] proved that every Jordan derivation of a prime ring of characteristic not 2 is a derivation.

In a braided monoidal category the braiding c replaces the swap of vector spaces

CONTACT A. Adrabi. Email: abderrahim.adrabi@um5r.ac.ma

[13, Definition 2.1]. The product $\mu(1+c)$ is the sum of the product and its braided opposite. Graziani et al. [8, Claim 2.14] state that, for a symmetric algebra (A, μ, u, c) in their sense (a braided unital algebra with $c^2 = \mathrm{id}_A$), the morphism $\mu \circ (\mathrm{Id}_{A\otimes A} - c)$ defines a braided Lie algebra structure on A; the claim is recalled there from Goyvaerts and Vercruysse. Pareigis [15, Definition 5.5] defines derivations of μ in the category of Yetter–Drinfeld modules. We found no source that treats derivations of $\mu(1+c)$ for a general braided monoidal category.

This paper is the third of a series on symmetrized products. The first paper defines braided-symmetric derivations of an algebra, which are derivations of the product $\mu + \mu\sigma$ for an involution σ of $A \otimes A$ [1, Definition 2.1]. The second treats the Lie derivations of an associative algebra as the derivations of its antisymmetrized product [2, Section 4]. A Lie derivation of an associative algebra is a linear map D with $D([x,y]) = [D(x), y] + [x, D(y)]$ [14, p. 183]. Here the category is general, and the proofs use the axioms of a braided monoidal category and the additivity of morphisms, together with, where a statement says so, symmetry, 2-torsion freeness, or closedness and kernels.

Section 2 recalls the notions of the sources. Section 3 proves that the Leibniz defect decides the derivations of $\mu(1 \pm c)$ (Proposition 3.2) and that a derivation of both products is a derivation of μ (Theorem 3.4). Section 4 treats a commutative product (Theorem 4.1) and the derivation object (Theorem 4.3). Section 5 twists the braiding by a morphism α (Theorem 5.1). We end with one open question.

2. Preliminaries

This section recalls the notions of the sources that the paper uses; apart from the specialisation to symmetric monoidal categories recorded below, it contains no new results.

Throughout, $\mathcal{C}$ is a braided monoidal category enriched over abelian groups: every hom-set is an abelian group, composition is bilinear, and the tensor product of morphisms is bilinear. Write $c_{X,Y} \colon X \otimes Y \to Y \otimes X$ for the braiding. We omit the isomorphisms of associativity and of the unit, by the coherence theorem for braided monoidal categories [13, Theorem 2.5]. Let $\mu \colon A \otimes A \to A$ be a morphism, and write $c = c_{A,A}$. We do not assume that μ is associative. Every statement below holds, in particular, in a symmetric monoidal category, in which $c_{Y,X} \circ c_{X,Y} = \mathrm{id}$ for all X, Y; the hypotheses that go beyond braidedness are named in the statements that use them, namely 2-torsion freeness in Theorems 3.4, 4.1 and 5.1(iii), and closedness with kernels in Theorem 4.3. There the braiding on $A \otimes A$ is an involution, so that $\mu(1 \pm c) = \mu \pm \mu c$ and

$$(1+c)(1-c) = (1-c)(1+c) = 0,$$

and the twist $\sigma = (\alpha \otimes \alpha)c$ of Section 5 is an involution whenever α is.

We recall Pareigis' definition of a derivation, which is the model for the notion below. Here C is a category of Yetter–Drinfeld modules.

Definition 2.1 ([15, Definition 5.5])**.** Let A be an algebra in C and let $\mathrm{end}(A)$ be the inner endomorphism object of A in C, that is, the Yetter–Drinfeld module $\mathrm{end}(A)$ satisfying $\mathsf{C}(X \otimes A, A) \cong \mathsf{C}(X, \mathrm{end}(A))$ for all $X \in \mathsf{C}$. His description of it is $\mathrm{end}(A) = \{f \in \mathrm{Hom}(A,A) \mid \exists \sum f_{(0)} \otimes f_{(1)} \in \mathrm{Hom}(A,A) \otimes K \ \forall a \in A : \sum f_{(0)}(a) \otimes f_{(1)} = \sum f(a_{(0)})_{(0)} \otimes f(a_{(0)})_{(1)} S(a_{(1)})\}$, which is the Yetter–Drinfeld module with the required

universal property. The object $\mathrm{end}(A)$ operates on A by a canonical map $\mathrm{ev}\colon \mathrm{end}(A)\otimes A \to A$ with $\mathrm{ev}(f\otimes a) = f(a)$. A derivation from A to A is a linear map $d\colon A\to A$ in $\mathrm{end}(A)$ such that $d(ab) = d(a)b + (1\otimes d)(\tau\otimes 1)(d\otimes a\otimes b)$ for all $a, b \in A$. Here K is the Hopf algebra whose Yetter–Drinfeld modules form C, S is its antipode, and $\sum f_{(0)}\otimes f_{(1)}$, $\sum a_{(0)}\otimes a_{(1)}$ are Sweedler's notation for the components of f and of the coaction on A. Equivalently, as Pareigis [15, proof of Corollary 5.6] puts it, d is a derivation when $m\circ(d\otimes 1 + 1\otimes d) = d\circ m$ for the product m of A, the tensor of $\mathrm{end}(A)$ with A being the braided one, $(x\otimes y)(a\otimes b) = (\mathrm{ev}\otimes\mathrm{ev})(1\otimes\tau\otimes 1)(x\otimes y\otimes a\otimes b)$. Observe that in the symmetric situation this means $d(ab) = d(a)b + ad(b)$.

For the unit object as parameter, the braiding τ in Definition 2.1 disappears, and the rule becomes a square.

Definition 2.2. Let $\nu\colon A\otimes A\to A$ be a morphism. A morphism $D\colon A\to A$ is a *derivation of* ν if $D\circ\nu = \nu\circ(D\otimes 1 + 1\otimes D)$.

Put $\mu_\pm = \mu\circ(1\pm c)$. We call μ_+ the *symmetrized product* and μ_- the *antisymmetrized product* of μ. Ion [10, Definition 1.5] defines a braided commutator on a braided algebra, in the sense of his Definition 1.4.

Definition 2.3 ([10, Definition 1.5])**.** Let $(A, \nabla, 1, c)$ be a braided algebra. The map $[\cdot,\cdot]_A\colon A^{\otimes 2}\to A$, $[\cdot,\cdot]_A = \nabla(\mathrm{id}_{A^{\otimes 2}} - c)$ is called the *braided commutator* on A.

When $(A, \mu, 1, c)$ is a braided algebra in the sense of Ion [10, Definition 1.4], the antisymmetrized product μ_- is the braided commutator of Definition 2.3. The rule of Definition 2.2 for μ_+ says that the following square commutes.

$$\begin{array}{ccc} A\otimes A & \xrightarrow{\ \mu_+\ } & A \\ {\scriptstyle D\otimes 1+1\otimes D}\downarrow & & \downarrow{\scriptstyle D} \\ A\otimes A & \xrightarrow[\ \mu_+\ ]{} & A \end{array}$$

For vector spaces with the swap, μ_+ is $x\otimes y\mapsto xy + yx$. Its derivations are the Jordan derivations of Herstein [9, p. 1104], as shown in the first paper of the series [1, Remark 2.4].

The defect below measures how far D is from a derivation of μ.

Definition 2.4 ([1, Definition 3.1])**.** The Leibniz defect of a linear map $D\colon A\to A$ is the bilinear map $E_D\colon A\times A\to A$ given, for $x, y\in A$, by $E_D(x,y) = D(xy) - D(x)y - xD(y)$.

3. The Leibniz defect

The map is classical: Cusack [6, Section 2] writes it $d(a,b) = D(ab) - aDb - (Da)b$ and studies Jordan derivations through it. The same formula gives the defect of a morphism $D\colon A\to A$ of $\mathcal{C}$, with $\Delta_D = D\otimes 1 + 1\otimes D$:

$$E_D = D\circ\mu - \mu\circ\Delta_D\colon A\otimes A\to A.$$

So D is a derivation of μ exactly when $E_D = 0$. The braiding commutes with Δ_D.

Lemma 3.1. $c \circ \Delta_D = \Delta_D \circ c$.

Proof. Since c is natural, $c \circ (D \otimes 1) = (1 \otimes D) \circ c$ and $c \circ (1 \otimes D) = (D \otimes 1) \circ c$. Adding the two equations gives the claim. □

The lemma is the commutative square

$$\begin{array}{ccc} A \otimes A & \xrightarrow{\ c\ } & A \otimes A \\ {\scriptstyle \Delta_D}\downarrow & & \downarrow{\scriptstyle \Delta_D} \\ A \otimes A & \xrightarrow[\ c\ ]{} & A \otimes A \end{array}$$

and it gives the main criterion.

Proposition 3.2. *Let $D\colon A \to A$ be a morphism. Then*

$$D \circ \mu_\pm - \mu_\pm \circ \Delta_D = E_D \circ (1 \pm c).$$

So D is a derivation of μ_+ exactly when $E_D \circ (1 + c) = 0$, and a derivation of μ_- exactly when $E_D \circ (1 - c) = 0$.

Proof. By Lemma 3.1, $\Delta_D \circ (1 \pm c) = (1 \pm c) \circ \Delta_D$. Hence

$$D\mu(1 \pm c) - \mu(1 \pm c)\Delta_D = D\mu(1 \pm c) - \mu\Delta_D(1 \pm c) = E_D(1 \pm c). \qquad \square$$

A derivation of μ has zero defect, so it satisfies both criteria.

Corollary 3.3. *A derivation of μ is a derivation of μ_+ and of μ_-.*

Proof. If $E_D = 0$, then $E_D \circ (1 \pm c) = 0$, and Proposition 3.2 applies. □

The converse needs one condition on $\mathcal{C}$. We say $\mathcal{C}$ is 2-*torsion free* if $f + f = 0$ implies $f = 0$ for every morphism f.

Cusack proved that every Jordan derivation of a 2-torsion free semiprime ring is a derivation [6, Theorem 4 and Corollary 5], and Brešar gave a short proof [4, Theorem 1]. The next theorem uses two products and no hypothesis on A.

Theorem 3.4. *Let $\mathcal{C}$ be 2-torsion free, and let $D\colon A \to A$ be a morphism that is a derivation of μ_+ and of μ_-. Then $E_D = 0$, that is, D is a derivation of μ.*

Proof. By Proposition 3.2, $E_D(1+c) = 0$ and $E_D(1-c) = 0$. Adding gives $E_D + E_D = 0$, so $E_D = 0$. □

4. The commutative case and the derivation object

Call μ *commutative* if $\mu \circ c = \mu$, the condition that Ion [10, Definition 1.4] calls c-commutative for a braided algebra.

Theorem 3.4 needs both products. For a commutative product the symmetrized product alone decides.

Theorem 4.1. *Let $\mathcal{C}$ be 2-torsion free and let μ be commutative. Then the derivations of μ_+ are the derivations of μ.*

Proof. By Lemma 3.1 and commutativity,

$$E_D c = D\mu c - \mu \Delta_D c = D\mu - \mu c \Delta_D = D\mu - \mu \Delta_D = E_D.$$

So $E_D(1+c) = E_D + E_D$. By Proposition 3.2, D derives μ_+ exactly when $E_D + E_D = 0$, that is, when $E_D = 0$. □

The proof is the triangle

$$\begin{array}{ccc} A \otimes A & \xrightarrow{\;c\;} & A \otimes A \\ & \searrow^{E_D} & \downarrow{\scriptstyle E_D} \\ & & A \end{array}$$

which commutes for a commutative product. For vector spaces with the swap, $\mu_+(x \otimes y) = 2xy$ for a commutative algebra. The theorem then says that every Jordan derivation of a commutative algebra is a derivation when no nonzero element is killed by 2.

The derivations of Pareigis form an object, not only a set. We show that the same holds for μ_+ when the braiding is a symmetry, that is, $c_{Y,X} \circ c_{X,Y} = \mathrm{id}$. For an object T, a *T-family* is a morphism $f \colon T \otimes A \to A$. Put

$$\ell_f = f \otimes 1, \qquad r_f = (1 \otimes f) \circ (c_{T,A} \otimes 1),$$

as morphisms $T \otimes A \otimes A \to A \otimes A$. Define the defect of the family by

$$E_f = f \circ (T \otimes \mu) - \mu \circ \ell_f - \mu \circ r_f \colon T \otimes A \otimes A \to A.$$

For the unit object T, the family f is a morphism $A \to A$, and E_f is the defect of the previous section. For Yetter–Drinfeld modules and the universal family $T = \mathrm{end}(A)$, $f = \mathrm{ev}$, the equation $E_f = 0$ is the rule of Definition 2.1; for the unit object the braiding disappears and the rule is the untwisted Leibniz rule. Write E_f^+ for the same expression with μ replaced by μ_+.

Lemma 4.2. *Let $\mathcal{C}$ be symmetric. For every T-family f,*

$$E_f^+ = E_f \circ (T \otimes (1 + c)).$$

Proof. We first show

$$c \circ \ell_f = r_f \circ (T \otimes c), \qquad c \circ r_f = \ell_f \circ (T \otimes c).$$

The first equation follows because c is natural in its first variable, applied to f, and from the hexagon axiom for $c_{T \otimes A, A}$. The second equation follows in the same way from the second variable and the hexagon axiom for $c_{A, T \otimes A}$. It also uses $c_{A,T} \circ c_{T,A} = \mathrm{id}$, which is where the symmetry is needed. Now

$$\mu_+ \ell_f + \mu_+ r_f = \mu \ell_f + \mu r_f (T \otimes c) + \mu r_f + \mu \ell_f (T \otimes c) = \mu(\ell_f + r_f)\,(T \otimes (1 + c)).$$

Also $f \circ (T \otimes \mu_+) = f \circ (T \otimes \mu) \circ (T \otimes (1 + c))$. Subtracting gives the claim. □

The lemma shows that the family defect for μ_+ is the family defect for μ composed with $T \otimes (1 + c)$. This gives the objects. Here *closed* means that each $- \otimes X$ has a right adjoint $[X, -]$, so that $\mathrm{Hom}(T, [X, Y]) \cong \mathrm{Hom}(T \otimes X, Y)$.

Theorem 4.3. *Let $\mathcal{C}$ be symmetric and closed, and let $\mathcal{C}$ have kernels. Then there are objects* $\mathrm{Der}(\mu)$ *and* $\mathrm{Der}(\mu_+)$ *with*

$$\mathrm{Hom}(T, \mathrm{Der}(\mu)) \cong \{f \colon T \otimes A \to A \mid E_f = 0\},$$
$$\mathrm{Hom}(T, \mathrm{Der}(\mu_+)) \cong \{f \colon T \otimes A \to A \mid E_f^+ = 0\},$$

natural in T, and $\mathrm{Der}(\mu)$ *is a subobject of* $\mathrm{Der}(\mu_+)$.

Proof. Write $[X, Y]$ for the internal hom, with $\mathrm{Hom}(T, [X, Y]) \cong \mathrm{Hom}(T \otimes X, Y)$. The assignment $f \mapsto E_f$ is natural in T, because the braiding is natural and the tensor product is functorial in each variable. A natural map between the functors represented by $[A, A]$ and $[A \otimes A, A]$ comes from a morphism, so $f \mapsto E_f$ is induced by a morphism $\Phi \colon [A, A] \to [A \otimes A, A]$. By Lemma 4.2, the assignment $f \mapsto E_f^+$ comes from $\Phi_+ = (1 + c)^* \circ \Phi$, where $(1 + c)^*$ is composition with $1 + c$. Put $\mathrm{Der}(\mu) = \ker \Phi$ and $\mathrm{Der}(\mu_+) = \ker \Phi_+$. The two isomorphisms follow from the universal property of the kernel and the defining property of the internal hom. Since $\Phi_+ = (1 + c)^* \Phi$, the kernel of Φ lies in the kernel of Φ_+. □

The kernel diagram of the proof is

$$\begin{array}{ccccc}
\mathrm{Der}(\mu) & \hookrightarrow & [A, A] & \xrightarrow{\Phi} & [A \otimes A, A] \\
\downarrow & & \downarrow{\scriptstyle \mathrm{id}} & & \downarrow{\scriptstyle (1+c)^*} \\
\mathrm{Der}(\mu_+) & \hookrightarrow & [A, A] & \xrightarrow[\Phi_+]{} & [A \otimes A, A]
\end{array}$$

and $\mathrm{Der}(\mu_+) = \ker \Phi_+$ contains $\mathrm{Der}(\mu)$. Only Lemma 4.2 uses the symmetry: the morphism Φ, and with it the object $\mathrm{Der}(\mu)$, is available already in a braided monoidal closed category in which Φ has a kernel, because the naturality of $f \mapsto E_f$ in T uses only the naturality of the braiding and the bifunctoriality of $\otimes$. The first object is the one of Pareigis.

Remark 4.4. For Yetter–Drinfeld modules, with no symmetry assumed, $\mathrm{Der}(\mu)$ is the object $\mathrm{Der}(A)$ of Pareigis [15, Definition 5.5 and the paragraph following it]: by the proof of his Corollary 5.6, a map $d \in \mathrm{end}(A)$ is a derivation exactly when $\mathrm{m}(d \otimes 1 + 1 \otimes d) = d\,\mathrm{m}$, an equation whose tensor is the braided one of that proof, and this is $\Phi(d) = 0$.

Barnes, Schenkel and Szabo define derivations as an equalizer in the module category of a quasi-Hopf algebra [3, Definition 3.4].

Definition 4.5 ([3, Definition 3.4])**.** Let H be a quasitriangular quasi-Hopf algebra and let A be a braided commutative algebra in the module category of H. The *derivations* of A is the object $\mathrm{der}(A)$ in that module category which is defined by the equalizer

$$\mathrm{der}(A) \longrightarrow \mathrm{end}(A) \rightrightarrows \mathrm{hom}(A, \mathrm{end}(A))$$

of the two morphisms $\zeta([\cdot,\cdot]\circ(\mathrm{id}\otimes\hat{l}))$ and $\zeta(\hat{l}\circ\mathrm{ev})$, where $[\cdot,\cdot]$ is the internal commutator of $\mathrm{end}(A)$, and $\hat{l} = \zeta(l)\colon A \to \mathrm{end}(A)$ is the currying of the left action $l\colon A\otimes A \to A$ induced by the product of A.

Theorem 4.3 gives $\mathrm{Der}(\mu)$ as a kernel in any symmetric closed category with kernels, and the definition above gives $\mathrm{der}(A)$ as an equalizer in the module category of a quasitriangular quasi-Hopf algebra.

5. A twisted involution

Let $\alpha\colon A \to A$ be a morphism and put $\sigma = (\alpha\otimes\alpha)\circ c$ and $\mu_\sigma = \mu\circ(1+\sigma)$. Since $\sigma^2 = (\alpha^2\otimes\alpha^2)\circ c^2$, the map σ is an involution of $A\otimes A$ when $\alpha^2 = \mathrm{id}_A$ and $c^2 = \mathrm{id}_{A\otimes A}$, in particular in a symmetric category; it is a braiding only when α is compatible with the braided structure, which we do not assume. For vector spaces with the swap and an involutive automorphism or anti-automorphism α, the map σ is the twisted swap of the first paper. There the derivations of μ_σ that commute with α equal the derivations of μ that commute with α, for a commutative reduced algebra with 2 invertible [1, Theorem 4.1]. We prove the part of that theorem that holds in a general category. We do not assume that α is compatible with the product. Fuchs, Runkel and Schweigert use an endomorphism $\varsigma\colon A\to A$ with $\varsigma\circ\mu = \mu\circ c\circ(\varsigma\otimes\varsigma)$ [7, Definition 2]. Their ς is a map on A, not on $A\otimes A$ like the σ above. We do not use that rule.

Theorem 5.1. *Let $D\colon A\to A$ satisfy $D\alpha = \alpha D$.*

(i) $D\mu_\sigma - \mu_\sigma\Delta_D = E_D\circ(1+\sigma)$.
(ii) If μ is commutative and D is a derivation of μ_σ, then $E_D + E_D\circ(\alpha\otimes\alpha) = 0$.
(iii) Let $\varepsilon\in\{1,-1\}$ and let $p,p'\colon X\to A$ satisfy $\alpha p = \varepsilon p$ and $\alpha p' = \varepsilon p'$. If $\mathcal{C}$ is 2-torsion free, μ is commutative and D is a derivation of μ_σ, then $E_D\circ(p\otimes p') = 0$.

Proof. (i) Since $D\alpha = \alpha D$, we have $\Delta_D(\alpha\otimes\alpha) = (\alpha\otimes\alpha)\Delta_D$. With Lemma 3.1, $\Delta_D\sigma = \sigma\Delta_D$. The computation of Proposition 3.2 gives (i).

(ii) By (i) and the hypothesis, $E_D(1+\sigma) = 0$. Now $E_D\sigma = E_D(\alpha\otimes\alpha)c$, and c commutes with $\alpha\otimes\alpha$ because c is natural. As in the proof of Theorem 4.1, $E_Dc = E_D$, so $E_D\sigma = E_D(\alpha\otimes\alpha)$. Hence $E_D + E_D(\alpha\otimes\alpha) = 0$.

(iii) Since $(\alpha\otimes\alpha)(p\otimes p') = \varepsilon^2(p\otimes p') = p\otimes p'$, composing (ii) with $p\otimes p'$ gives $E_D(p\otimes p') + E_D(p\otimes p') = 0$. So $E_D(p\otimes p') = 0$. □

For vector spaces with 2 invertible and $\alpha^2 = \mathrm{id}_A$, put $A_\pm = \ker(\alpha\mp 1)$; then $A = A_+\oplus A_-$. Part (iii) says that E_D vanishes on $A_+\otimes A_+$ and on $A_-\otimes A_-$. The vanishing on $A_+\otimes A_-$ does not follow from (iii): it needs a reduced commutative algebra [1, Theorem 4.1].

We end with a question. In the survey of Brešar and Zelmanov, algebras are associative and over a field of characteristic not 2 [5, Section 1 and Section 11], and the following theorem holds. An algebra is *semiprime* if $aAa = 0$ implies $a = 0$.

Theorem 5.2 ([5, Theorem 11.1])**.** *A Jordan derivation of a semiprime associative algebra over a field of characteristic not* 2 *is a derivation.*

The theorem is due to Cusack [6, Corollary 5], as the survey notes.

For vector spaces with the flip as braiding, Corollary 3.3 gives $\mathrm{Der}(\mu)\subseteq\mathrm{Der}(\mu_+)$, and Theorem 5.2 gives the converse inclusion for semiprime associative algebras. Which

hypotheses on $(\mathcal{C}, A)$ give $\mathrm{Der}(\mu_+) = \mathrm{Der}(\mu)$ when μ is not commutative?

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